

Deep Learning Architectures for Document AI

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1. Introduction

Document understanding bridges visual perception and linguistic semantics. Traditional optical character recognition systems primarily emphasize character transcription, often discarding vital spatial relationships.

Modern document intelligence frameworks integrate vision backbones with graph neural representations and multimodal transformers to preserve hierarchical reading flows across complex multi-column structures.

2. Methodology

We propose a two-stage topological ordering algorithm. In the first phase, bounding polygons are projected along vertical density axes to isolate distinct layout columns. In the second phase, reading flow graphs enforce strict reading order.

By preventing horizontal cross-column drift, the extracted text maintains semantic integrity for downstream large language model retrieval.